

POLYP DETECTION MEDICAL REPORT

Analysis Date: June 04, 2026

VIDEO INFORMATION

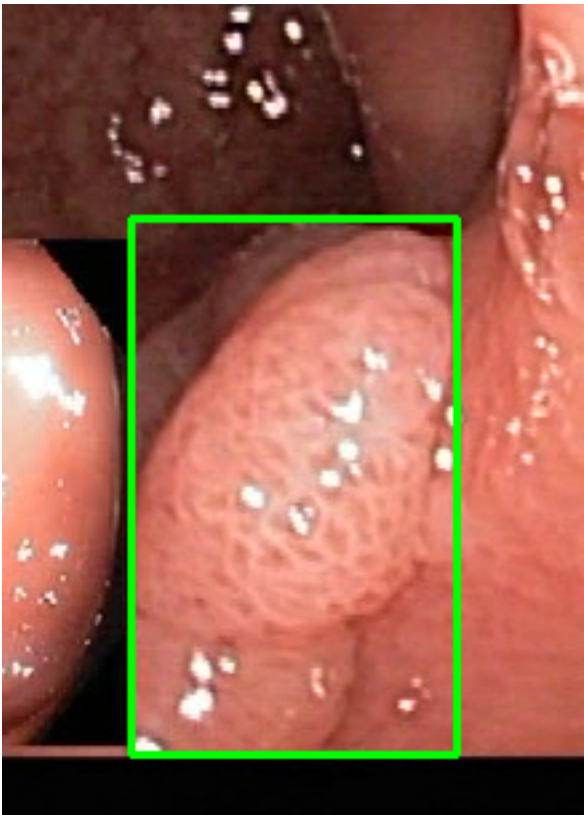
Video ID:	f75949a3-bd10-4762-a7d5-1e5e42fae8e4
Total Frames:	1683
Frame Rate:	25.0 fps (assumed)
Duration:	67.3 seconds

DETECTION SUMMARY

Detection Method	Count	Status
YOLO Detections	1830	✓
RT-DETR Detections	2003	✓
MedSAM2 Detections	2208	✓
Consensus (3-Model Agreement)	4	✓
Risk Classification	Count	Percentage
HIGH RISK	1	25.0%
MEDIUM RISK	0	0.0%
LOW RISK	3	75.0%
TOTAL ANALYZED	4	100.0%

DETAILED POLYP ANALYSIS

POLYP #1 — LIFTED_POLYP [LOW_RISK] (Tracked across 49 frames)



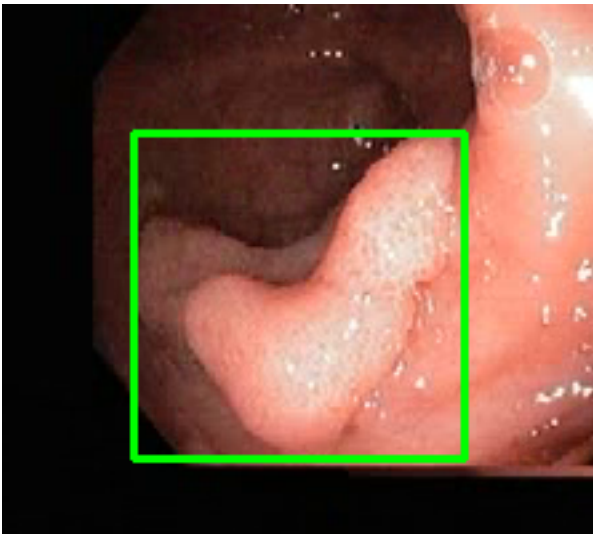
Feature	Value	Interpretation
Frame Index	1256	Frame 1256 of 1683
Detection Method	Detector Consensus (MedSAM2 Unavailable)	Detector Consensus (MedSAM2 Unavailable) (MedSAM2 unavailable for this video)
POLYP TYPE	LIFTED_POLYP	Type confidence: 52.1%
Redness Score	0.031	Color-based vascularization
Relative Radius	12.5%	Polyp size as % of frame diagonal
Texture Score	0.031	Surface morphology
Vessel Visibility	0.000	Vascularization indicator
Risk Tier	LOW RISK	Validated by ESGE 2017 / NICE classification
Clinical Classification	LOW_RISK	Final symbolic reasoning bucket

Expert Prediction	LOW_RISK	Decision Tree output
Clinical Class	NORMAL_MUC OSA	Normal mucosal pattern — no polyp morphology identified
Detection Confidence	87.7%	YOLO / RT-DETR / track confidence

Type Assessment: Lifted lesion post-injection — submucosal plane confirmed. Resection feasible.

Low-confidence LOW RISK detection - Follow standard protocol

POLYP #2 — LIFTED_POLYP [LOW_RISK] (Tracked across 948 frames)



Feature	Value	Interpretation
Frame Index	1256	Frame 1256 of 1683
Detection Method	Detector Consensus (YOLO / RT-DETR / MedSAM2 Unavailable)	Detector Consensus (YOLO / RT-DETR / MedSAM2 Unavailable) (MedSAM2 unavailable for this video)
POLYP TYPE	LIFTED_POLYP	Type confidence: 52.1%
Redness Score	0.031	Color-based vascularization
Relative Radius	12.5%	Polyp size as % of frame diagonal
Texture Score	0.031	Surface morphology
Vessel Visibility	0.000	Vascularization indicator
Risk Tier	LOW RISK	Validated by ESGE 2017 / NICE classification
Clinical Classification	LOW_RISK	Final symbolic reasoning bucket

Expert Prediction	LOW_RISK	Decision Tree output
Clinical Class	NORMAL_MUC OSA	Normal mucosal pattern — no polyp morphology identified
Detection Confidence	86.0%	YOLO / RT-DETR / track confidence

Type Assessment: Lifted lesion post-injection — submucosal plane confirmed. Resection feasible.

Low-confidence LOW RISK detection - Follow standard protocol

POLYP #3 — LIFTED_POLYP [LOW_RISK] (Tracked across 299 frames)



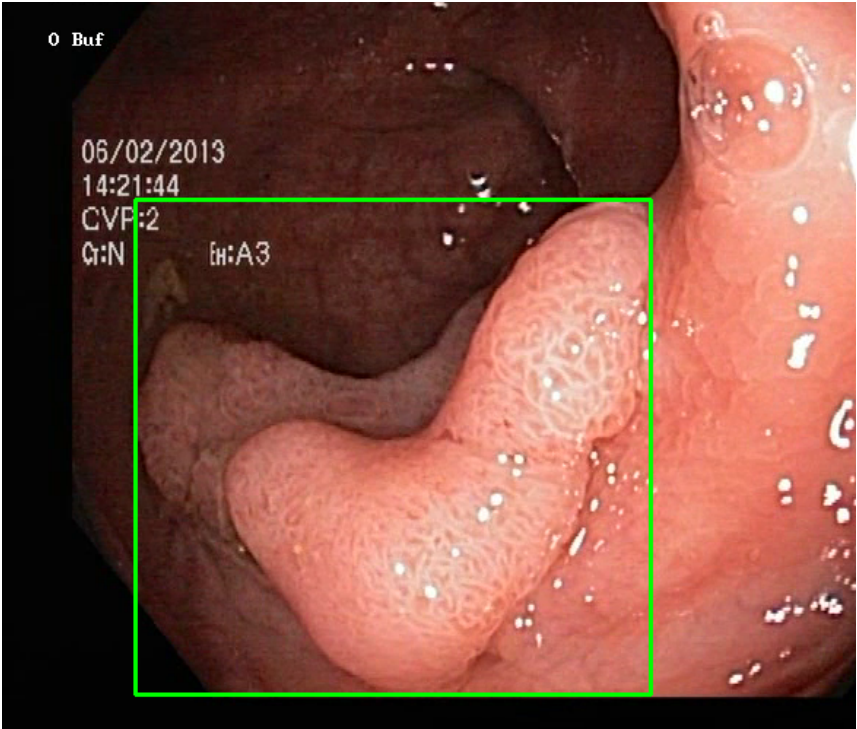
Feature	Value	Interpretation
Frame Index	1256	Frame 1256 of 1683
Detection Method	Detector Consensus (YOLO / RT-DETR / MedSAM2 Unavailable)	MedSAM2 Unavailable (MedSAM2 unavailable for this video)
POLYP TYPE	LIFTED_POLYP	Type confidence: 52.1%
Redness Score	0.031	Color-based vascularization

Relative Radius	12.5%	Polyp size as % of frame diagonal
Texture Score	0.031	Surface morphology
Vessel Visibility	0.000	Vascularization indicator
Risk Tier	LOW RISK	Validated by ESGE 2017 / NICE classification
Clinical Classification	LOW_RISK	Final symbolic reasoning bucket
Expert Prediction	LOW_RISK	Decision Tree output
Clinical Class	NORMAL_MUC OSA	Normal mucosal pattern — no polyp morphology identified
Detection Confidence	84.0%	YOLO / RT-DETR / track confidence

Type Assessment: Lifted lesion post-injection — submucosal plane confirmed. Resection feasible.

Low-confidence LOW RISK detection - Follow standard protocol

POLYP #4 — BLEEDING_POLYP [HIGH_RISK] (Tracked across 56 frames)



Feature	Value	Interpretation
Frame Index	1176	Frame 1176 of 1683
Detection Method	Detector Consensus (YOLO / RT-DETR)	Model Agreement (MedSAM2 unavailable for this video)
POLYP TYPE	BLEEDING_POLYP	Type confidence: 92.0%
Redness Score	0.222	Color-based vascularization
Relative Radius	43.7%	Polyp size as % of frame diagonal
Texture Score	0.280	Surface morphology
Vessel Visibility	0.773	Vascularization indicator
Risk Tier	HIGH RISK	Validated by ESGE 2017 / NICE classification
Clinical Classification	HIGH_RISK	Final symbolic reasoning bucket
Expert Prediction	HIGH_RISK	Decision Tree output
Clinical Class	MALIGNANT_POLYP	High suspicion for malignancy — biopsy and urgent resection recommended
Detection Confidence	89.7%	YOLO / RT-DETR / track confidence

Type Assessment: Active hemorrhagic lesion — elevated vascularity with redness signal. Immediate clinical review required.

Low-confidence HIGH RISK detection - Further evaluation recommended

CLINICAL IMPRESSION

Summary:

A total of 4 polyps were detected and analyzed using multi-model consensus and AI-based risk stratification.

Risk Distribution:

- HIGH RISK polyps: 1 (25.0%)
- MEDIUM RISK polyps: 0 (0.0%)
- LOW RISK polyps: 3 (75.0%)

Recommendation:

The presence of HIGH RISK polyps warrants close clinical attention. Consider endoscopic resection or follow-up surveillance depending on clinical context.

Technical Details:

- Detection: Multi-model consensus (YOLO, RT-DETR, MedSAM2)
- Features: 444-dimensional vectors (384 SSL + 60 biomarkers)
- Classification: Cluster-specific Decision Tree experts
- Confidence: Based on deep learning + medical heuristics

Disclaimer: This report is generated by an AI-assisted research system for academic and research purposes only. All clinical recommendations are derived from published ESGE 2017, NICE CG118/CG131, and BSG 2019 colonoscopy guidelines. This output does not constitute medical advice and must not be used as the sole basis for clinical decisions. All findings must be reviewed and confirmed by a qualified gastroenterologist or endoscopist.